

HYDROINFORMATICS

DATA MANAGEMENT AND ANALYSIS

Graduate Course

Lecture 3: Introduction to the course

Learning Outcomes

By the end of the course, students will be able to

1. Explain the fundamental concepts, principles, and applications of Hydroinformatics in water resources management.
2. Acquire, manage, and preprocess hydrological and geospatial datasets from ground-based and Earth observation sources.
3. Analyze and visualize hydrological data using statistical methods and computational tools to identify trends, patterns, and anomalies.
4. Apply Python and cloud-based geospatial platforms to solve common water resources problems involving floods, droughts, and water quality.
5. Interpret and communicate analytical results to support evidence-based decision-making in water resources management.

The background of the slide is a vibrant blue gradient. Overlaid on this are numerous 3D-rendered numbers in white and light blue. These numbers, including digits 0 through 9, are of various sizes and are scattered across the frame, creating a sense of depth and movement. Some numbers are in the foreground, appearing larger and more detailed, while others are in the background, smaller and more faded.

Introduction

Hydroinformatics

HYDRO-INFORMATICS: An Applied Science

Hydro = Water

Informatics = Study of Computational Systems

Hydroinformatics is about information management to aid in understanding and managing water in the natural and built environments.

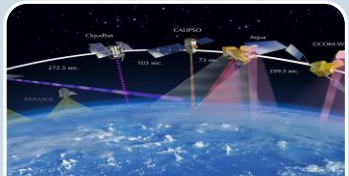
Computer Science

Geographic Information System

Hydrology

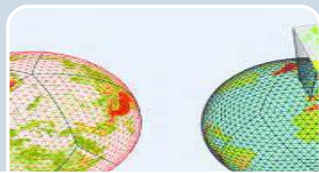
Engineering

High End Computing



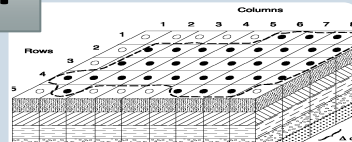
Data

- Earth Observation
- Ground Data



Models

- Numerical Model
- Hydrologic Models
- Hydraulic Models
- AI Models



Knowledge

Weather Patterns/
Predictions
Environmental
condition
Climate change impact



Visualization

Maps, graphs
showing
relationships and
trends, etc.



Decision Support

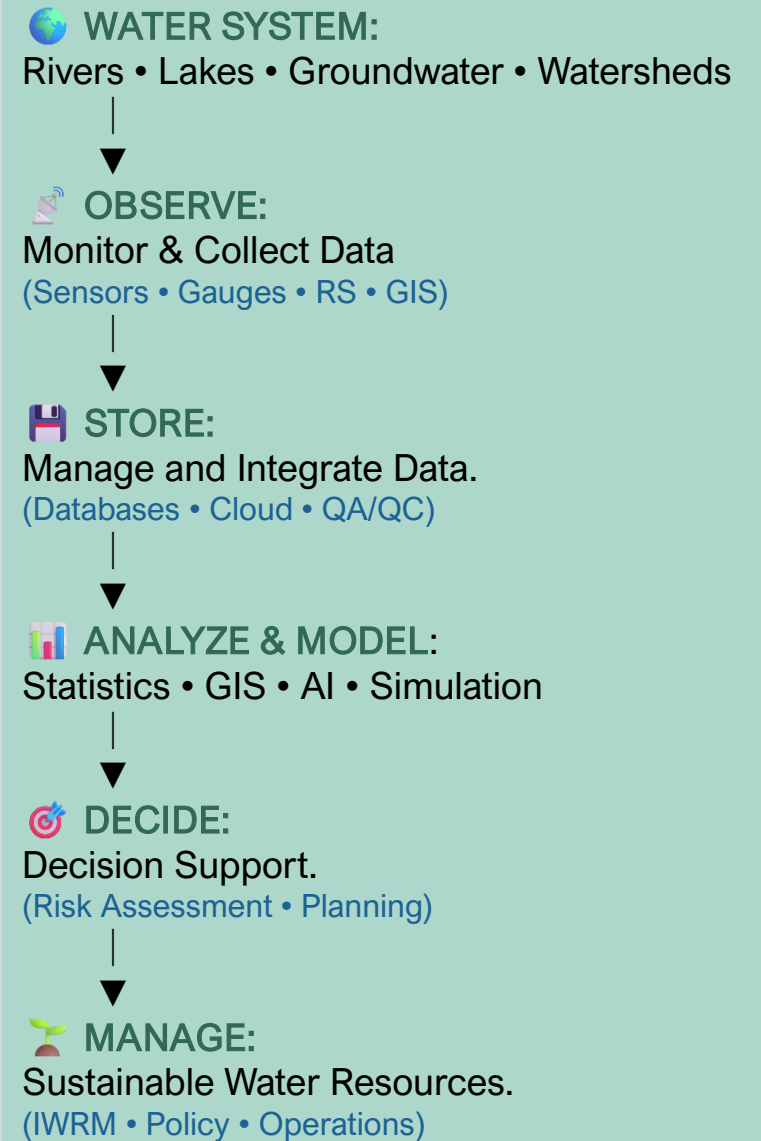
Environmental,
economic, and
social impacts of
different scenarios

DEFINITION

- ❖ Hydroinformatics encompasses the effective use of data for the characterization of the water cycle through natural and built environments.
- ❖ It involves development of theories, methodologies, algorithms and tools, methods for testing the concepts, analysis and verifications, and knowledge representation and communication.

Why is Hydroinformatics considered an interdisciplinary field?

Hydroinformatics combines hydrology, data science, and computing to support integrated water resources management (IWRM).



Core Components



Hydrology & Hydraulics



Remote Sensing & GIS



Data Science, Statistics, & Artificial Intelligence



Numerical & Hydraulic Modeling



IoT, Sensors & Real-time Monitoring



Cloud Computing & Decision Support Systems

Why Hydroinformatics?

- ◈ Integrates diverse water-related data.
- ◈ Supports evidence-based decision making.
- ◈ Improves monitoring, forecasting, and early warning
- ◈ Enables sustainable water resources management
- ◈ Helps address climate change and water security challenges



Applications

1. Flood forecasting and early warning
2. Drought monitoring
3. Water quality assessment
4. Irrigation and agriculture Urban water management
5. Groundwater monitoring
6. Climate change impact assessment

Some Examples

Digital Twin



- ◆ A Digital Twin is a digital model of a real-world system that is continuously updated using real-time or near-real-time data to monitor, understand, and predict its behavior.

Example in Hydroinformatics:

- ◆ A digital twin of a river system can combine river gauge data, satellite observations, rainfall, weather forecasts, and hydrological models to show the current river condition and predict future water levels or floods.



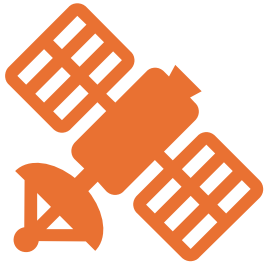
Flood Forecasting & Early Warning

Live Hydrograph Feeds: Stream tracking charts showing real-time water volume changes.

River Gauge Indicators: Color-coded alert bars changing from green to flashing red.

GIS Inundation Maps: Highlighted layout zones indicating active flood boundaries.

Drought Monitoring



Soil Moisture Mapping: High-contrast satellite layouts tracking ground dryness trends.



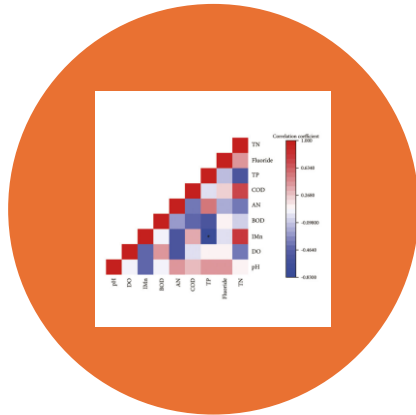
Vegetation Stress Indices: Color gradients charting declining crop health.



Precipitation Deficit Charts: Timeline bar graphs comparing current rainfall against averages.



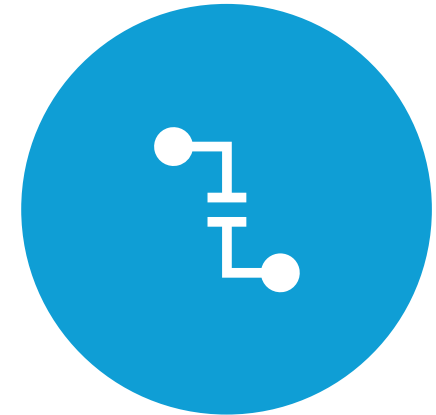
Water Quality Assessment



CONTAMINATION HEATMAPS: VISUAL
CHEMICAL DISPERSION MODELS
ACROSS ACTIVE RESERVOIRS.



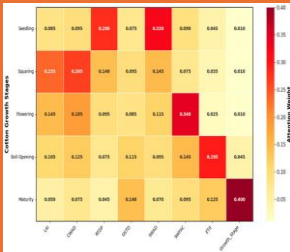
KPI MATRIX DASHBOARD: RING
METERS TRACKING PH, DISSOLVED
OXYGEN, AND TURBIDITY.



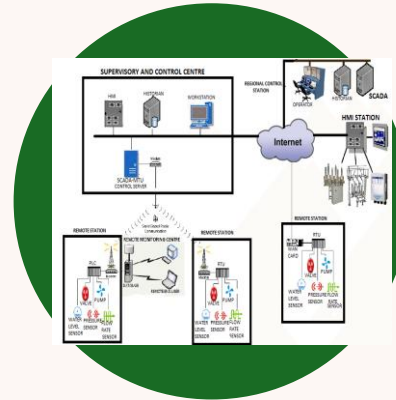
SENSOR MESH GRIDS: NODE
LAYOUTS SHOWING PLACEMENT OF
LIVE AUTOMATED SAMPLING SITES.



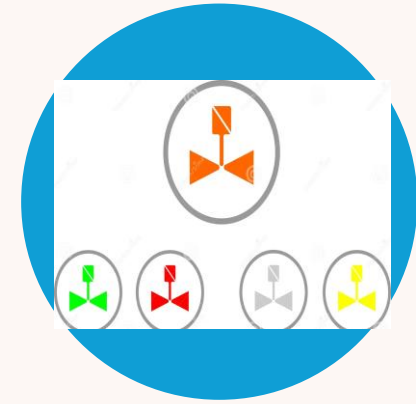
Irrigation & Agriculture



CROP WATER DEMAND: HEATMAPS TRACKING FIELDS REQUIRING IMMEDIATE WATER DELIVERY.



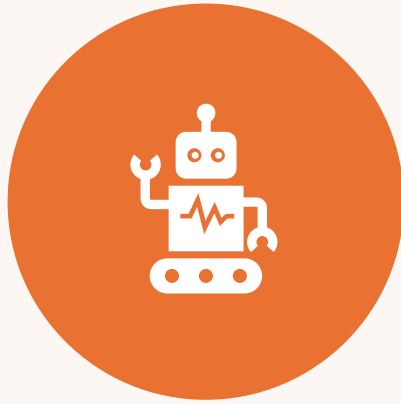
SCADA CANAL SCHEMES: LIVE FLOW SCHEMATICS TRACKING PRIMARY DISTRIBUTION CANALS.



AUTOMATION VALVE STATUS: SIMPLE TOGGLE ICONS SHOWING ACTIVE OPEN OR CLOSED PIPES.



Urban Water Management



SMART UTILITY TWIN: A DIGITAL 3D MODEL TRACKING THE MUNICIPAL PIPE NETWORK.



ACOUSTIC LEAK ALERTS: PULSING RADAR SYMBOLS ISOLATING PIPELINE STRUCTURAL BREAKS.



DEMAND PEAK GAUGES: VERTICAL BAR CHARTS MONITORING CITY-WIDE WATER CONSUMPTION.

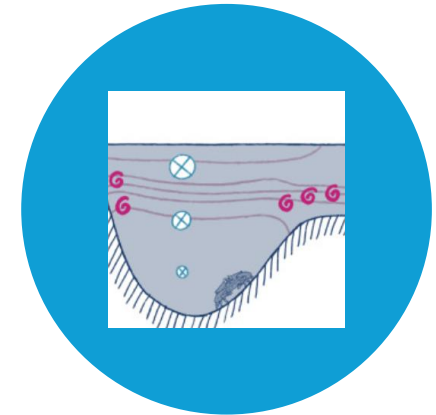
Groundwater Monitoring



BOREHOLE DEPTH STATUS: VERTICAL
CROSS-SECTION GRAPHICS
TRACKING AQUIFER LEVELS.

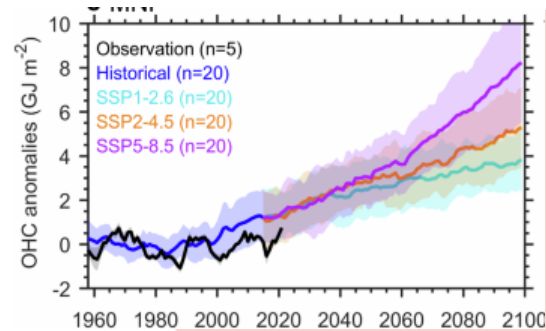


RECHARGE VS. DRAWDOWN: SPLIT
COLUMN CHARTS DISPLAYING NET
WATER VOLUME SHIFTS.

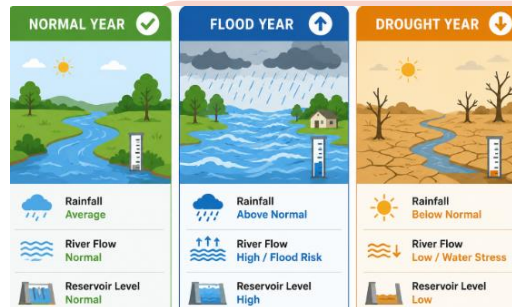


SALINITY INTRUSION VECTORS:
LINEAR GRADIENT ARROWS POINTING
OUT COASTAL SEAWATER MIXING.

Climate Change Impact



Decadal Trend Lines: Multi-colored trajectory lines tracking long-term warming.

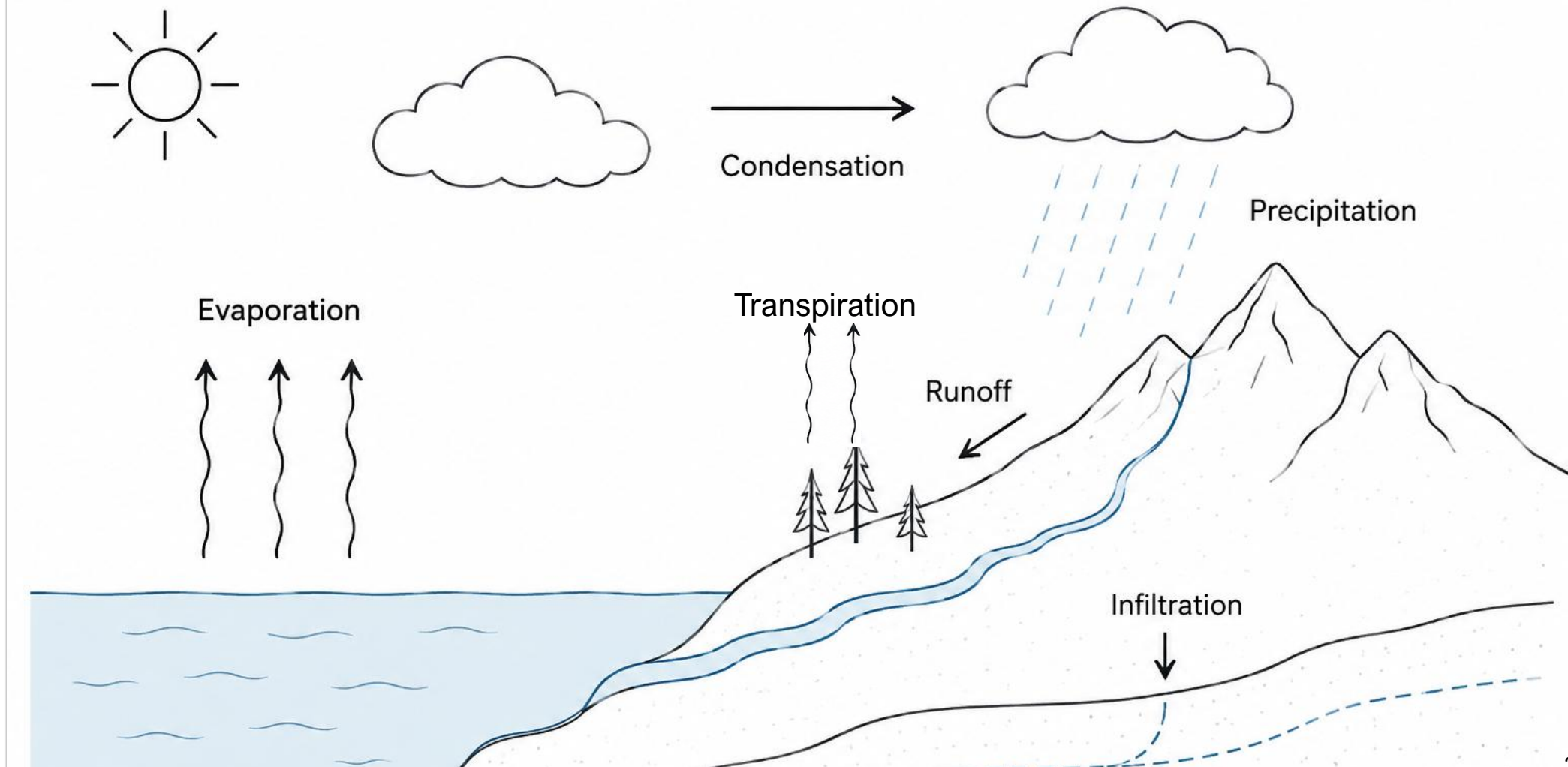


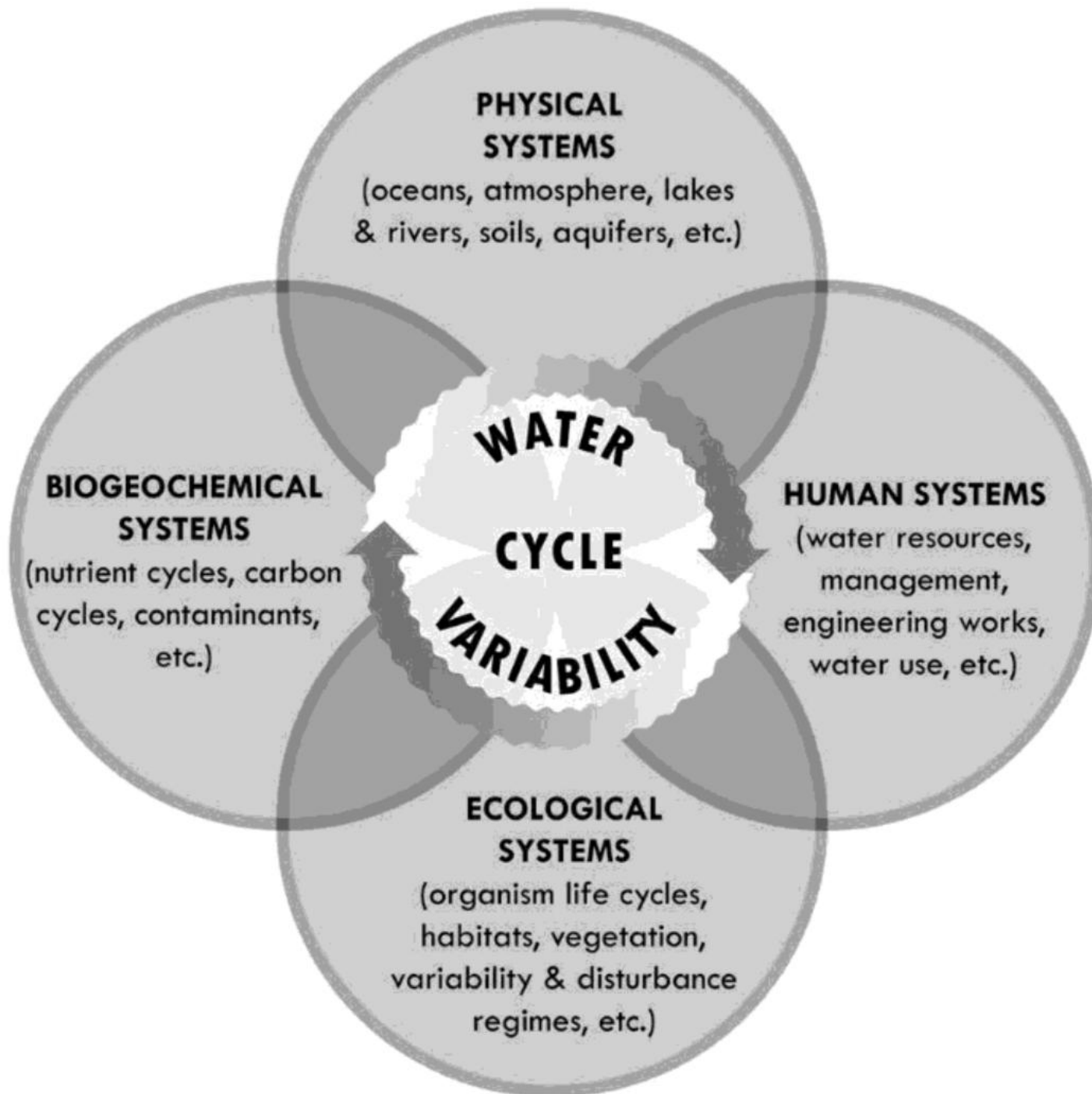
Scenario Matrix Blocks: Side-by-side comparison tiles simulating extreme weather.



Glacial Melt Timelines: Graphic progress bars visualizing ice reserve losses.

THE WATER CYCLE



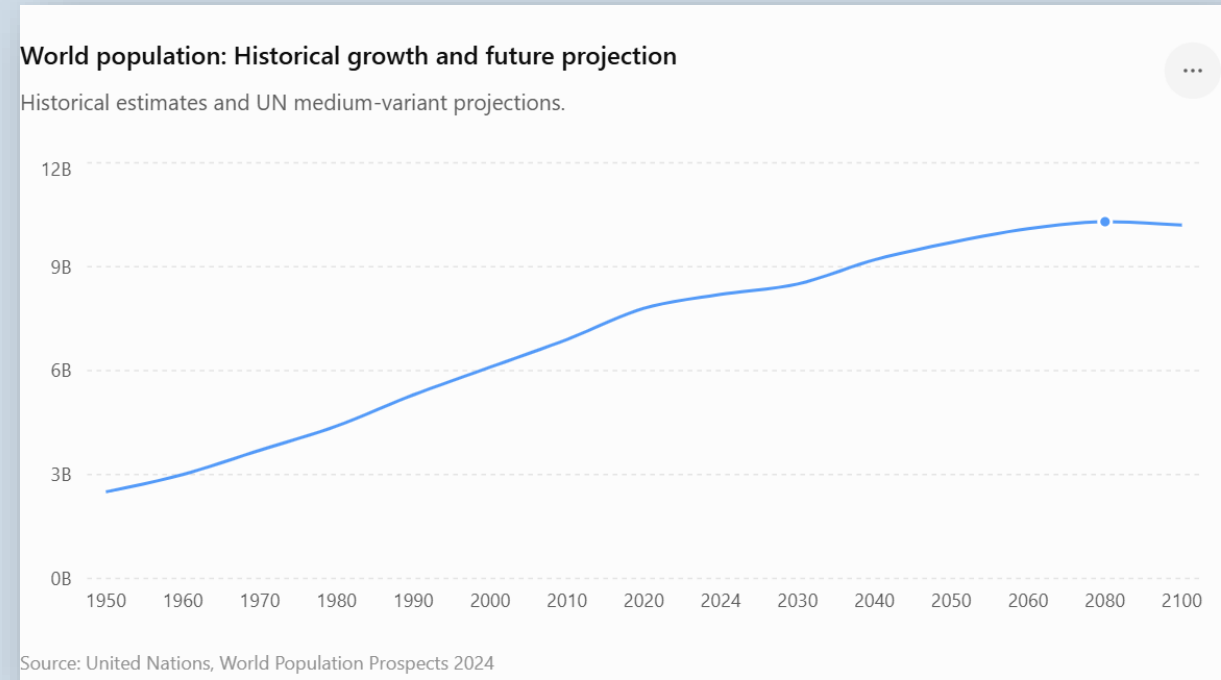


Global Water
Cycle is a
Complex
System!!!

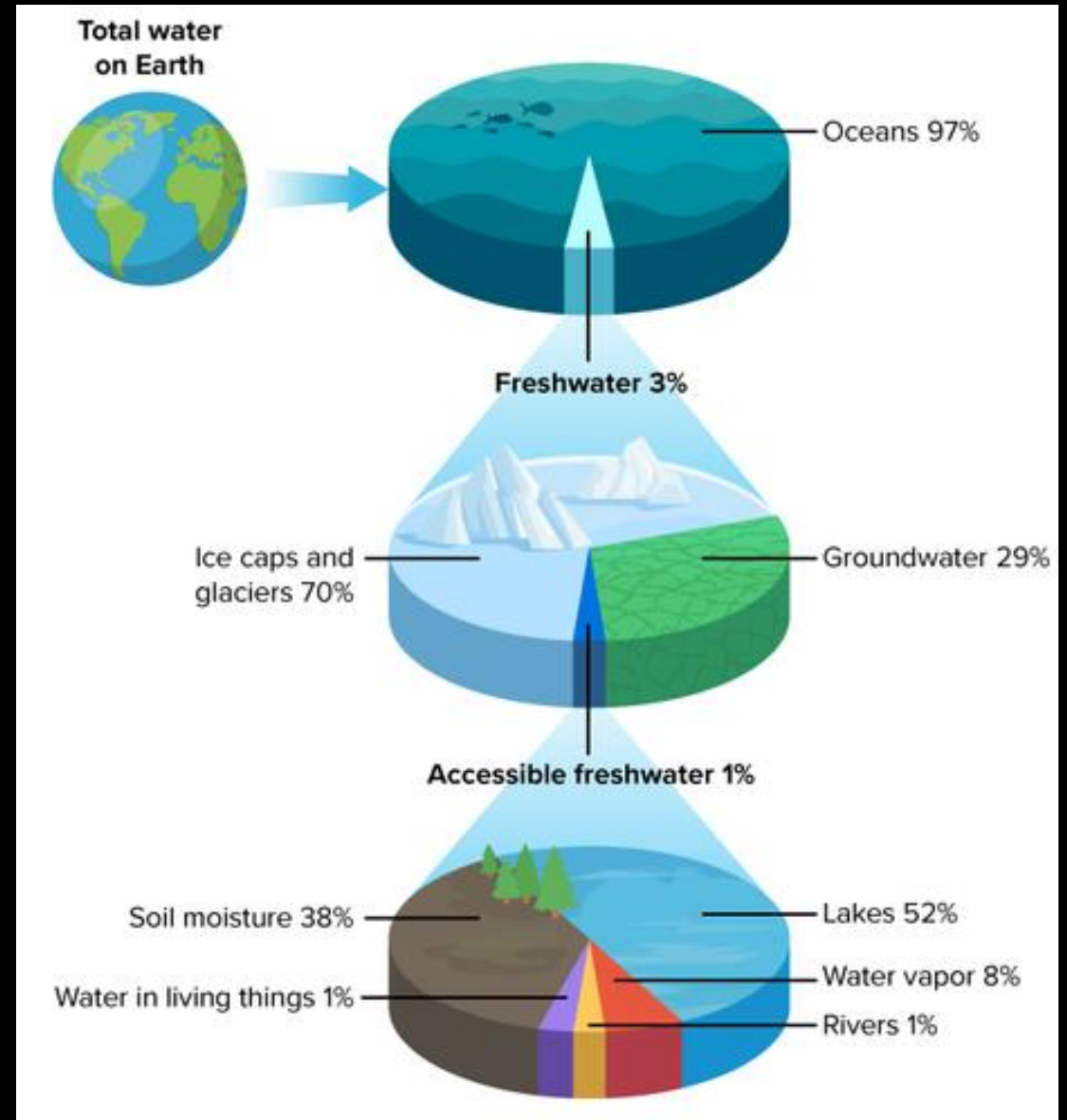
Water Management

Discuss!!!

1. What are the causes of water scarcity and their impacts?
2. How to optimize?
3. How to minimize uncertainty?



Global Water Distribution



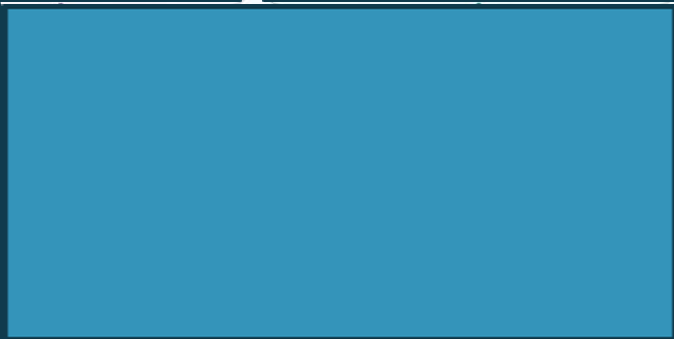
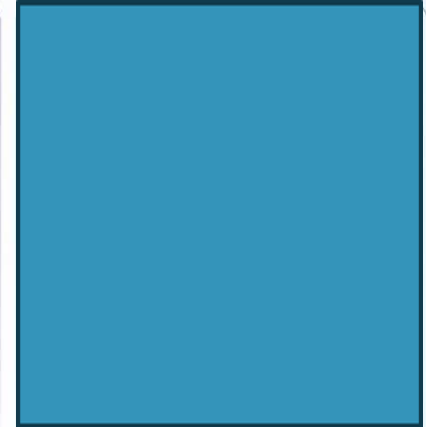
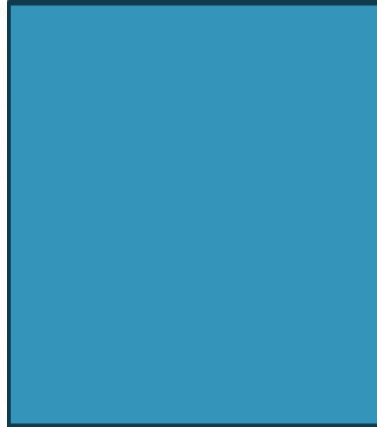
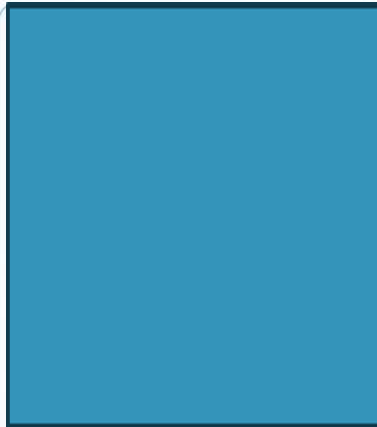
DRIVERS OF CHANGE AND THEIR IMPACT ON GLOBAL WATER AVAILABILITY

Global water availability is under increasing pressure from multiple, interlinked factors. These drivers not only affect the quantity, but also the quality and accessibility of freshwater.

1. CLIMATE CHANGE



- Rising temperatures increase evapotranspiration.
- Altered precipitation patterns lead to droughts and floods.
- Melting glaciers reduce long-term freshwater storage.



WATER IS A FINITE RESOURCE.

Understanding these drivers and taking collective action today is essential to secure water for people and the planet tomorrow.

EVERY DROP COUNTS.

Data Requirement

Interactive classroom activity

Imagine you are responsible for managing a river basin. What data would you collect to make informed water management decisions? Classify your answers into:

1. Hydro-meteorological datasets
2. Geospatial datasets

Data Sources

Interactive classroom activity

Can you identify some commonly used hydro-meteorological and geospatial datasets? Give a few examples.

Hydro-meteorological and Geospatial Data Sources

Ground-Based Observations

 **Weather data:** rainfall, temperature, humidity, wind speed, solar radiation (e.g., Pakistan Meteorological Department - PMD)(e.g., PMD)

 **Hydrological data:** river discharge, water levels, canal flows, reservoir storage((e.g., Irrigation Department, WAPDA, IRSA)e.g., Irrigation Department, WAPDA, IRSA)

 **Groundwater data:** groundwater levels and well observations

 **Water quality data:** pH, EC, TDS, dissolved oxygen, turbidity, etc.

Earth Observation (Satellite Data)

 **Precipitation:** CHIRPS, GPM, etc.

 **Vegetation:** NDVI, EVI. Etc.

 **Evapotranspiration:** WaPOR, MODIS ET, etc.

 **Land Use/Land Cover:** Sentinel-2, Landsat, ESRI Land Cover

 **Surface Water & Floods:** Sentinel-1/2, Global Surface Water

 **Water Levels:** Satellite Altimetry (Sentinel-3, ICESat-2, Jason-3)

 **Groundwater Storage:** GRACE

GIS & Remote Sensing

- ◆ Applications in water resources

Hydrological Models

- ◆ Conceptual, physical and data-driven

Decision Support

- ◆ Forecasting, planning and management

Discussion Stop #1

- ◆ Why is traditional hydrology alone no longer sufficient?

Before You Leave Today

1. One new thing learned?
2. What remains unclear?